

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: ML A03
COURSE CODE: Reinforcement learning for
Environment and Agent

COURSE OUTCOME COVERED

COT: Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems (BL2,BL3)

Assessment Tool Weightage: 30%

Dr Dumpala Prasanth

QUESTIONS: 10
Marks: 50

Total

Annexure A

Case Study
Duration: 2hrs

1. A logistics company is developing an RL-based warehouse robot that learns to pick, pack, and transport items efficiently while avoiding obstacles and minimizing delays. Analyze how the components of a Markov Decision Process (states, actions, rewards, policy, and environment) can be defined for this system. Design a reward function that balances speed, accuracy, and safety, and evaluate how reward design influences the robot's learning and behavior.

(5 Marks)

2. A streaming platform wants to use Reinforcement Learning to recommend movies dynamically based on user preferences and viewing history. Apply RL

concepts to construct a suitable state representation, action space and reward function. Analyze how different reward signals affect user engagement and evaluate the trade-off between exploration and exploitation. **(5 Marks)**

3. An autonomous agricultural system uses RL to optimize irrigation and fertilizer usage based on soil conditions and weather forecasts. Design an RL framework by defining states, actions, and rewards. Analyze the effect of delayed rewards on crop yield optimization and justify the choice of a suitable learning algorithm. **(5 Marks)**
4. A cybersecurity system uses RL to detect and respond to network attacks in real time. Analyze how states and environment can be modeled for intrusion detection. Design appropriate actions and reward functions, and evaluate the challenges of sparse rewards and real-time decision-making. **(5 Marks)**
5. A ride-sharing company plans to use RL to optimize driver allocation and dynamic pricing strategies. Apply RL concepts to define states, actions, and rewards. Analyze how environmental uncertainty affects learning and evaluate ethical concerns related to pricing strategies. **(5 Marks)**
6. A smart home automation system uses RL to control lighting, heating, and cooling based on occupancy and weather conditions. Design an RL framework including states, actions, and reward function. Analyze how conflicting objectives like comfort and energy saving can be handled and evaluate the impact of reward shaping. **(5 Marks)**
7. A gaming company is developing an RL-based agent

to play a complex strategy game against human players. Analyze how the problem can be modeled as an MDP. Design a reward structure that encourages long-term strategy and evaluate how exploration strategies affect performance. (5 Marks)

8. A financial institution uses RL to detect fraudulent transactions in real time. Apply RL principles to define the state space, action space, and reward function. Analyze the challenges of imbalanced data and delayed feedback, and evaluate the risks associated with incorrect decisions. (5 Marks)

9. A healthcare system uses RL to recommend personalized rehabilitation exercises for patients based on their progress. Design an RL framework with suitable states, actions, and rewards. Analyze the impact of delayed rewards and patient variability, and evaluate ethical concerns and safety considerations. (5 Marks)

10. A smart grid system uses RL to manage electricity distribution and reduce peak load demand. Analyze how the components of an MDP can be defined in this context. Design a reward function to balance efficiency and reliability, and evaluate the challenges of scalability and real-time learning. (5 Marks)

Code of Conduct:

I R. Geethika (192425101) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R. Geethika

RUBRICS FOR CASE STUDY

Criteria	Weightage	Level 4 Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 Beginning
Understanding of Case	25	Thorough understanding clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Reinforcement learning using openAI

Problem Understanding

The objective is to develop a Reinforcement learning (RL) agent that learns the optimal path in a grid world environment.

The agent starts from an initial position and must reach the goal while avoiding the obstacles and minimizes unnecessary the Movements.

Constraints:-

- * The agent cannot Move outside the grid.
- * obstacle cells cannot be entered
- * The episode ends with the goal is reached are the maximum number of steps is exceeded.
- * The shortest safe path should be preferred.

State space:-

The state represents the current position of the agent in the grid.

Action space:-

The agent can perform four actions

- up
- down
- left
- right

Reward function:-

- * Goal reached +100
- * Normal movement: -1
- * Obstacle hit: -20
- * Invalid Move: -10

This reward function encourages the agent to reach the goal quickly while avoiding obstacles.

Constraint analysis

The learning algorithm must balance exploration and exploitation while ensuring the agent avoids restricted cells and follows a feasible path.

Justification:-

This design satisfies all constraints by enabling the agent to safely reach the destination with the minimum number of steps while maximizing cumulative reward.

2. ϵ -Greedy multi-armed Bandit Algorithm Problem Understanding

The objective is to maximize Total Reward using an ϵ -greedy multi-armed Bandit Algorithm under a limited budget of 100 trials.

Constraints:-

- * Total trials = 500
- * Only one arm is selected in each trial.
- * ϵ values considered.
 - 0.1
 - 0.3
 - 0.5

Constraints analysis

$$\epsilon = 0.1$$

- * Mostly exploits the best arm.
- least exploration

$$\epsilon = 0.5$$

- High exploration
- slower convergence.

Solution Development

The ϵ -greedy algorithm randomly explores with probability ϵ and exploits the best-known arm with probability $(1-\epsilon)$

Application of Concepts

- Exploration
- Exploitation
- Action value estimation
- Reward maximization.

Justification

An ϵ value of 0.3 provides the best balance between exploration under the 500-trial constraint.

3. Markov Decision process for Autonomous Robot

Problem Understanding:-

Design an MDP that enables an autonomous robot to reach a destination while minimizing energy consumption.

Constraints:-

- limited battery capacity.
- Avoid obstacles.
- Reach destination using minimum energy.
- Every movement consumes energy.

States:-

Each state consists of

- Robot position
- Remaining battery level.

Actions

- Move up
- Move down
- Move left

- Move right
- stay

Transition probabilities

- Intended movement: 90%.
- Movement failure: 10%.

This accounts for Uncertainty in robot motion.

Reward function

- Destination reached: -100
- Every movement: -2
- Obstacle collision: -50
- Battery exhausted: -100

Energy Constraints

- Each Move consumes One Energy Unit.
- Robot cannot continue when battery reaches zero.

Solution Development

The MDP computes the optimal policy using transition probabilities and expected future rewards.

Justification

The proposed MDP satisfies all constraints by ensuring safe navigation with minimum energy consumption.

4. Value Iteration for Constrained Grid Environment

Problem Understanding

The objective is to determine the optimal Policy in a constrained grid where certain states are restricted.

Constraints

- Restricted cells cannot be entered.
- Goal state must be reached.
- Invalid actions are penalized.

Constraint Analysis

The algorithm must ignore restricted states while maximizing long-term reward.

Bellman Equation

$$V(s) = \max_a [R(s, a) + \gamma \sum p(s'/s, a) V(s')]]$$

- $V(s)$ = value of state.
- $R(s, a)$ = immediate reward.
- γ = discount factor
- $p(s'/s, a)$ = Transition Probability.

Solution Development

Value iteration repeatedly updates state values using the Bellman Equation until

Convergence. The optimal action for each state forms the optimal policy.

Application of Concepts

- Dynamic programming
- Bellman Equation
- value iteration
- Optimal policy.

Justification

Value iteration guarantees convergence to the optimal policy while respecting all restricted state constraints.

5. RL framework for Warehouse Robot Problem Understanding

Design a Reinforcement learning framework that enables a warehouse robot with only 30 minutes of battery life to complete the max. number of delivery

Constraints

- * Battery capacity = 30 minutes.
- * Battery must not be exceeded.
- * Robot must avoid collisions.
- * Time should be minimized.

State space:-

Each state includes:-

- * Robot location
- * Delivery status
- * Remaining battery.

Actions:-

- * Move forward
- * Turn left
- * Turn Right
- * Pick item
- * Deliver item
- * Return to charging station.

Reward Design

- Successful delivery :- +100
- Efficient Movement :- +10
- Idle action :- 0
- Battery exhausted :- -100
- collision :- -50

Policy:-

The policy selects actions that Maximize deliveries while ensuring sufficient battery remains to complete tasks safely.

Constraint Analysis

Battery life is the primary constraint. The robot must prioritize nearby deliveries and avoid unnecessary movements.

Justification

The framework ensures the robot completes the maximum number of deliveries without exceeding the minimum battery limit, making it both efficient and practical.

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